



BCL-PSK

PSK MODULATION / DEMODULATION

KIT

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

In BCL-PSK kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATION

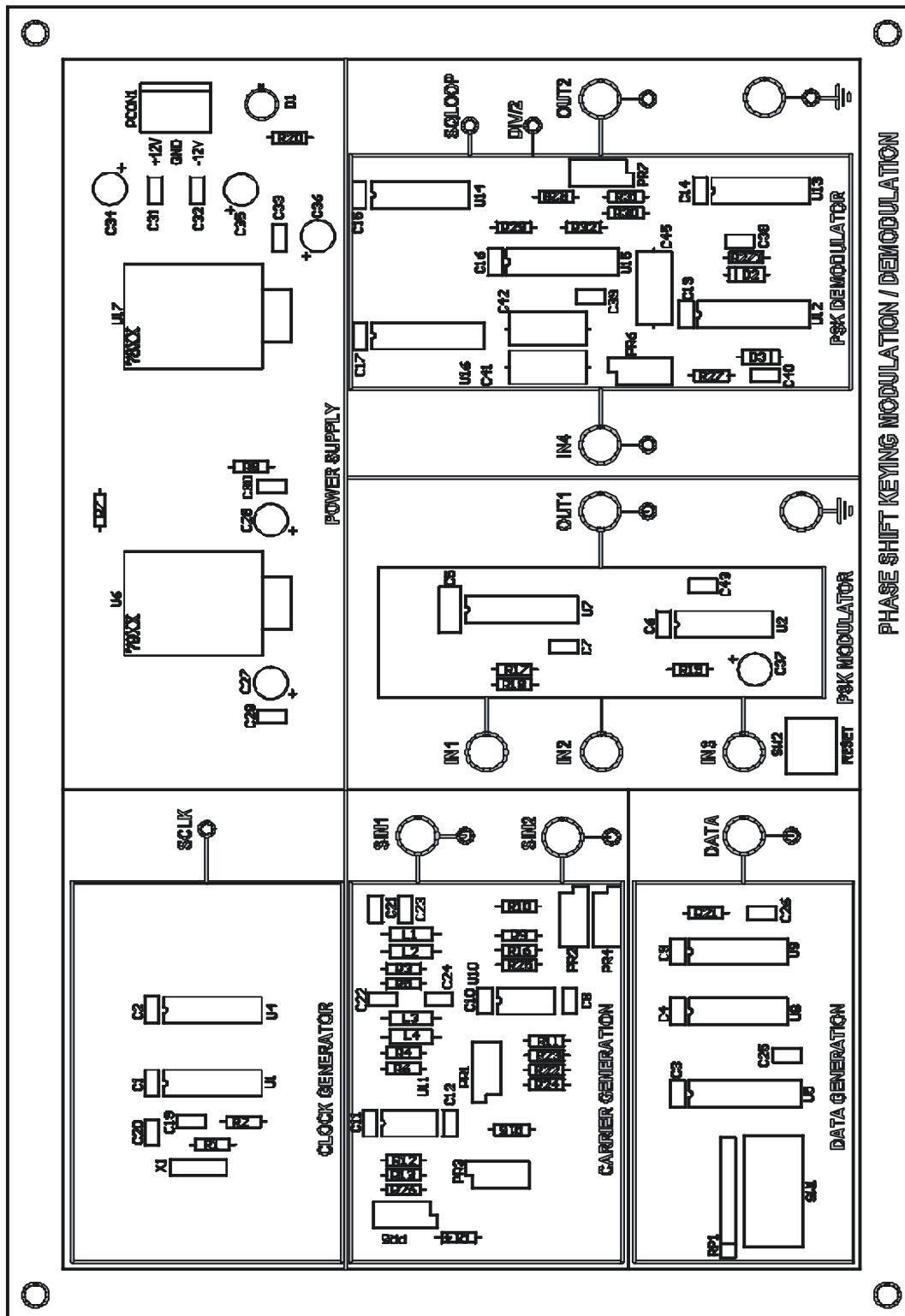
BCL-PSK: PHASE SHIFT KEYING MODULATION/DEMODULATION

DATA SIMULATOR	:	Onboard 8-bit variable NRZ-L pattern.
CRYSTAL OSCILLATOR	:	2 MHz
DATA CLOCK	:	250 KHz.
ON BOARD CARRIER SINE WAVES	:	1 MHz (0°) and 500KHz, 4VPP
CARRIER MODULATION	:	PSK.
CARRIER DEMODULATION	:	PSK
INTERMEDIATE SIGNAL	:	Provision for observing intermediate signals during demodulation.
INTER CONNECTION	:	2 mm banana socket.
POWER SUPPLY	:	+12V, -12V, GND.

ACCESSORIES

BCL-PSK		Quantity
1	Short Link Patch Cords male to male	4
2	12V Power Adapter	1
3	BCL-PSK EXPT Manual	1

FUNCTIONAL BLOCK



INTRODUCTION

“BCL-PSK” initiates the user to the various Carrier Modulation Techniques, normally adopted in practice

“BCL-PSK”: PHASE SHIFT KEYING MODULATOR / DEMODULATOR KIT.

FEATURES:

- 1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and the reference clock (256 KHz), which enables the trainers to work in a stand-alone mode.
- 2) Carrier modulation options are available on the kit: Phase Shift Keying (PSK)
- 3) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The three carrier sine waves generated onboard are of frequencies.
 - a) 1.024KHz (0°)
 - b) 512KHz (0°)
- 4) Interconnection facilities:
Sockets and patch chords are provided for on board connections.
- 5) Test Points:
All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.
- 6) Power Supply:
-12V, +12V, GND.

**EXPERIMENT
NO. 1**

EXPERIMENT NO 1

Aim:

To study the Phase Shift Keying Modulation and Demodulation.

Equipment:

PSK board, power supply, patch chord, Dual channel oscilloscope/DSO.

Theory:

Phase shift keying is another form of constant amplitude angle modulated digital modulation technique. PSK is similar to conventional PM except that the modulating signal is a binary signal and a limited number of output phases are possible.

With binary phase shift keying, two output phase changes are possible for a single carrier frequency. One output phase represents logic 1 and the other represents for logic 0. The carrier shifts between the two phase changes are 0 and 180°. Other names of BPSK are phase reversal keying and biphase modulation. BPSK is a form of suppressed-carrier, square-wave modulation of a continuous wave (CW) signal.

The input signal may be a $+\sin \omega_c t$ or $-\sin \omega_c t$. The coherent carrier recovery circuit detects and generates a carrier signal that is both frequency and phase coherent with the original transmit carrier. The balanced modulator produces the product of two inputs (the BPSK signal and recovered signal). The low pass filter separates the recovered binary data from the complex demodulated signal.

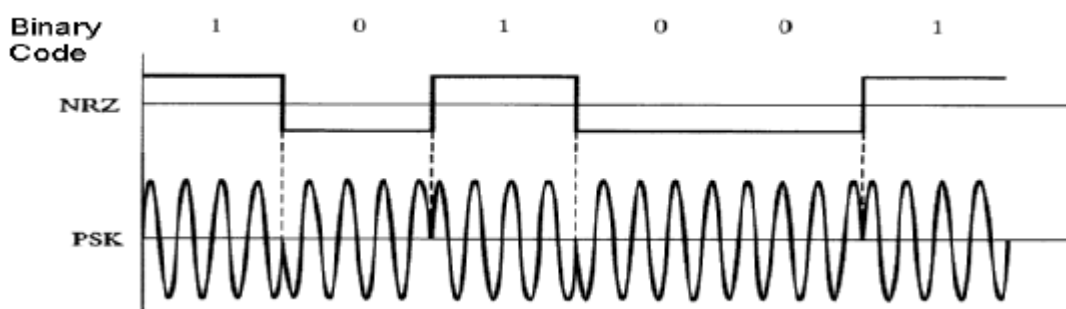


Fig 2. PSK waveforms

Applications of PSK:

Owing to PSK's simplicity, particularly when compared with its competitor quadrature amplitude modulation, it is widely used in existing technologies.

The wireless LAN standard, IEEE 802.11b-1999, uses a variety of different PSKs depending on the data rate required. At the basic rate of 1 Mbit/s, it uses DBPSK (differential BPSK). To provide the extended rate of 2 Mbit/s, DQPSK is used. In reaching 5.5 Mbit/s and the full rate of 11 Mbit/s, QPSK is employed, but has to be coupled with complementary code keying. The higher-speed wireless LAN standard, IEEE 802.11g-2003, has eight data rates: 6, 9, 12, 18, 24, 36, 48 and 54 Mbit/s. The 6 and 9 Mbit/s modes use OFDM modulation where each sub-carrier is BPSK modulated. The 12 and 18 Mbit/s modes use OFDM with QPSK. The fastest four modes use OFDM with forms of quadrature amplitude modulation.

Because of its simplicity BPSK is appropriate for low-cost passive transmitters, and is used in RFID standards such as ISO/IEC 14443 which has been adopted for biometric passports, credit cards such as American Express's Express Pay, and many other applications.

Bluetooth 2 will use $\pi/4$ -DQPSK at its lower rate (2 Mbit/s) and 8-DPSK at its higher rate (3 Mbit/s) when the link between the two devices is sufficiently robust. Bluetooth 1 modulates with Gaussian minimum-shift keying, a binary scheme, so either modulation choice in version 2 will yield a higher data-rate. A similar technology, IEEE 802.15.4 (the wireless standard used by ZigBee) also relies on PSK. IEEE 802.15.4 allows the use of two frequency bands: 868–915 MHz using BPSK and at 2.4 GHz using OQPSK.

Both QPSK and 8PSK are widely used in satellite broadcasting. QPSK is still widely used in the streaming of SD satellite channels and some HD channels. High definition programming is delivered almost exclusively in 8PSK due to the higher bitrates of HD video and the high cost of satellite bandwidth.^[5] The DVB-S2 standard requires support for both QPSK and 8PSK. The chipsets used in new satellite set top boxes, such as Broadcom's 7000 series support 8PSK and are backward compatible with the older standard

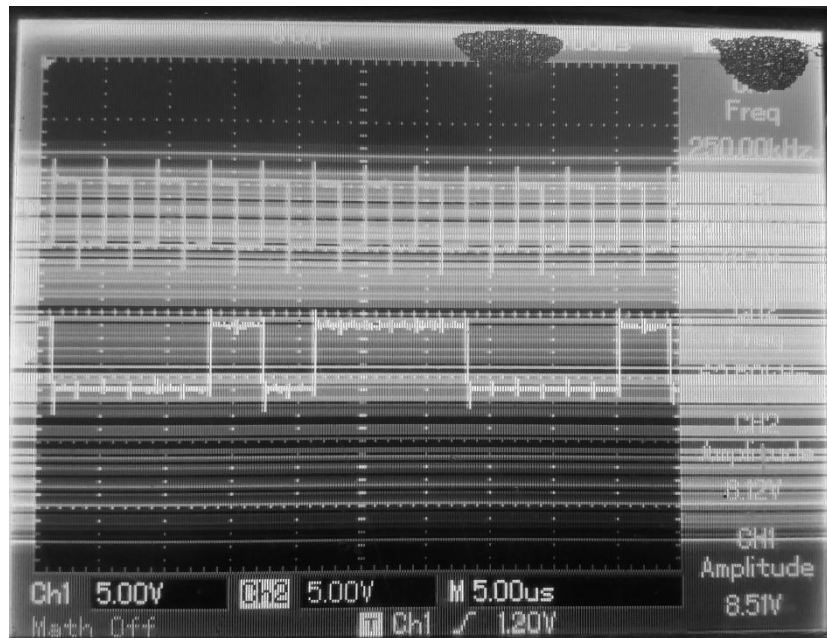
Procedure:

1. Connect power supply to kit with proper polarity and switch it ON.
2. Refer to KIT and carry out connections as mentioned below.
3. Observe carrier frequency at SIN1 and SIN2 posts and note down the same. Both signals are 180° out of phase.
4. Set data pattern as 10111001 using switch SW1 at DATA post.
5. Connect SIN1 post to IN2 post and SIN2 post to IN1 using patch chord.
6. Connect DATA post to IN3 post.
7. Observe PSK modulated signal at OUT1 post. Compare it with DATA. It is similar to fig 2 shown above.
8. To demodulate this signal connect OUT1 post to IN4 post.
9. PSK demodulator uses PLL detector technique to demodulate the PSK signal.

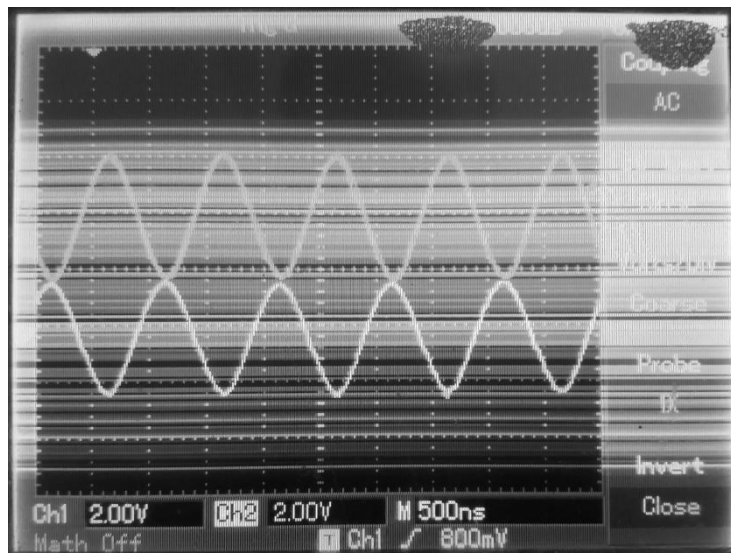
10. Observe recovered data pattern at OUT2 post. Compare it with DATA.

Captured Waveforms:

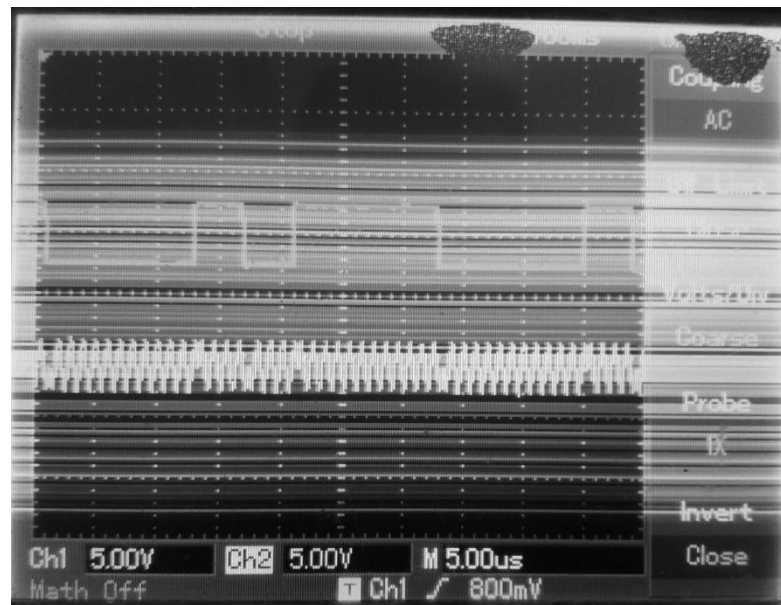
1) Clock and Data.



2) Carrier (1MHZ, 180° phase shifted)



3) PSK Modulation



4) PSK Demodulation

